

MIRRIAM CHISUNTA CHIKWAMU

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EDUCATION

Whitman College

Jan 2023 - Dec 2024

GPA: 3.79/4.0

Bachelor of Arts in Mathematics-Statistics; *Minor*: Economics

Relevant Coursework: Applied Machine Learning, Econometrics, Data Structures and Algorithms

Albertus Magnus College (Transferred)

Aug 2020 - Dec 2022

GPA: 3.96/4.0

Major: Mathematics

RELEVANT EXPERIENCE

Oak Ridge Institute for Science and Education Trainee | Navy Postgraduate School

Monterey, CA

July 2025 - Present

Symbolic AI Researcher

- Engineered formal logic representations by translating natural language into SUO-KIF axioms for SUMO ontology, with approved definitions integrated into SigmaKEE's live production instance
- Built Java-based proof translation system converting ATP outputs into Basic English, enabling non-technical military stakeholders to interpret formally verified reasoning
- Evaluated 400+ sentences for semantic validity to create labeled dataset for validating logical inference pipelines beyond surface-level grammar
- Integrated LLMs via Ollama to paraphrase ATP proofs while maintaining mathematical guarantees—unlike standalone LLM outputs, which hallucinate, providing formally verified reasoning with accessibility

Economic and Financial Statistics Division | Zambia Statistical Agency

Lusaka, Zambia

Aug 2024 - Sept 2024

Statistician Intern

- Identified and flagged 100+ pricing anomalies in 1,000+ data points through cross-referencing with historical baselines and domain knowledge of agricultural markets, preventing corrupted data from affecting Zambia's August 2024 national economic indicators
- Redesigned legacy digital forms using SQL/Access database principles, improving structural alignment with hard-copy surveys; increased throughput from 60 to 200 surveys/day (3.3x improvement), completing 4-week digitization project in 1 week

Department of Statistics & Data Science | Carnegie Mellon University

Pittsburgh, PA

June 2024 - July 2024

Sports Analytics Fellow | Advised by: Pittsburgh Penguins Research and Development Team

Project Title: Analyzing Consistency Among NHL Forwards' Performance

- Developed Weighted Mean Change framework using PCA-weighted z-scores to create combined consistency metric from EVG, EVA, and shooting percentage, validating year-to-year metric stability ($R^2 = 0.18-0.34$) across 200+ NHL forwards over 8 seasons in R
- Applied 4-quadrant clustering to categorize players by skill vs. consistency, revealing that teams prioritize performance over reliability in salary decisions—providing Penguins R&D with data-driven insights for roster evaluation—[Project Link](#)

Institute for Quantitative Biomedicine | Rutgers University

New Brunswick, NJ

June 2023 - Aug 2023

Rosetta Commons Research Intern | Khare Lab | Advised by: Sagar Khare, Vidur Samar

Project Title: Enhancing Site-Specific Cleavage by Benchmark Protease Enzymes through Bioinformatic Analysis of 3C Protease

- Performed computational analysis of 3C protease using MMseq2 for homology search across viral protein databases (~253M entries in UniProt), identifying catalytic residue sequence motifs through machine learning-based pattern recognition
- Discovered 5 promising enzyme variants for experimental validation and confirmed catalytic mechanisms from lab's preprint paper, enabling wet lab team to prioritize targeted therapeutic design candidates
- Synthesized computational insights into actionable reports for experimental researchers, bridging gap between bioinformatics predictions and practical enzyme engineering applications

Global Forest Watch with iXperience

Jan 2022 - Jan 2022

Data Analysis Remote Micro Internship

- Analyzed global deforestation patterns using Python and SQL, identifying top 10 countries with highest tree cover loss rates and visualizing trends with Seaborn and Matplotlib
- Synthesized findings into report with actionable conservation insights and policy recommendations

SKILLS

Programming Languages: Python, R, SQL, Java, C++, JavaScript, HTML/CSS

ML/AI Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, Random Forests, CNNs, ANNs

Knowledge Representation: SUMO, SUO-KIF, Automated Theorem Proving (E-Prover, Vampire), SigmaKEE

Data Science & Visualization: NumPy, pandas, Seaborn, Matplotlib, Plotly, tidyverse, ggplot2, dplyr, Mosiac, Quarto, RShiny

Statistical Methods: A/B Testing, ANOVA, Hypothesis Testing, Regression (Linear, Logistic, LASSO, Ridge), Clustering (k-Means, DBSCAN, Hierarchical), PCA, SVMs, k-NN, Decision Trees

NOTABLE PROJECTS

Senior Thesis: Predicting NHL Forward Successes Using Random Forests

Sep 2024–Dec 2024

- Built Random Forest models achieving 0.90 R^2 (90% variance explained) to predict NHL forward salary cap, contract length, and performance consistency, outperforming traditional ARIMA (0.60 R^2) by integrating Tango-adjusted metrics and custom variance calculations
- Designed interactive RShiny dashboard enabling stakeholders to compare algorithmic vs. traditional model predictions with dynamic visualizations of feature importance and performance trends—[Project Link](#)

Digital Heritage Archive Project (Hackathon)

April 2024 - April 2024

- Built interactive world map and cultural heritage recommendation system by fine-tuning Gemini Vision Pro 1.2 for image-based destination recognition, winning Whitman College's inaugural hackathon—[GitHub](#), [StreamLitApp](#)
- Led cross-functional team through project execution and presented strategic insights on AI adoption for tourism and cultural heritage preservation applications

LEADERSHIP, HONORS & AWARDS

Events Coordinator | Whitman African Student Association

Aug 2023 - Dec 2024

Scholarship | International Association of Black Actuaries

Aug 2023 - May 2024

Co-Founder | Chichasa NGO - Youth initiative in Zambia [Website link](#)

Oct 2019 - Present

Social media | HERZ Non-Governmental Org. [Website link](#)

Jan 2021 – Present